

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0510

Del Amo 138 kV
July 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0510. The Interconnection Customer proposes a 1009.0 MW natural gas combined cycle facility interconnecting at the Del Amo 138 kV substation in the Virginia area. The request was received on August 20, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,546,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Bull Run / Fillmore 138kV: 292.2% loading (323 MVA) following the outage of the Del Amo / Grand Blanc Generating Sta 138kV; DFAX = 100.0%.

Bull Run / Del Amo 138kV: 277.5% loading (348 MVA) following the outage of the Del Amo / Grand Blanc Generating Sta 138kV; DFAX = 100.0%.

Fillmore / S Houston 138kV L: 262.4% loading (236 MVA) following the outage of the Del Amo / Grand Blanc Generating Station Substation 138kV; DFAX = 58.2%.

The Del Amo / Grand Blanc Generating Station 138kV L is overloaded to 242.8% of its 398 MVA emergency rating for the loss of the Bull Run / Del Amo 138kV line. The project contributes 100.0% DFAX (1009.0 MW) to this overload.

The Castleton Energy Ctr / Sully Substation 138kV line is overloaded to 226.0% of its 365 MVA emergency rating for the loss of the Bull Run / Del Amo 138kV. The project contributes 59.2% DFAX (597.1 MW) to this overload.

The Sully / Walney 138kV line is overloaded to 216.8% of its 391 MVA emergency rating for the loss of the Bull Run / Del Amo 138kV. The project contributes 59.2% DFAX (597.1 MW) to this overload.

Grand Blanc Generating Sta / Walney 138kV: 204.4% loading (432 MVA) following the outage of the Bull Run / Del Amo 138kV line; DFAX = 59.2%.

For the loss of Grand Blanc Generating Sta / Del Amo 138kV Line, the S Houston / Dessau 138kV L loads to 188.1% of its 228 MVA rating. Project DFAX: 31.0%.

The Gallows Rd / Dessau 138kV L is overloaded to 185.4% of its 224 MVA emergency rating for the loss of the Del Amo / Grand Blanc Generating Station 138kV. The project contributes 31.0% DFAX (312.6 MW) to this overload.

For the loss of Walney / Grand Blanc Generating Station 138kV line, the Toa Vaca / Thomaston 138kV L loads to 129.9% of its 230 MVA rating. Project DFAX: 20.3%.

For the loss of Grand Blanc Generating Sta / Walney 138kV, the Fountain / Pleasant Vly Substation 138kV line loads to 127.7% of its 459 MVA rating. Project DFAX: 51.3%.

Pleasant Valley / Cub-Run Dp 138kV: 125.9% loading (481 MVA) following the outage of the Grand Blanc Generating Sta / Walney 138kV; DFAX = 51.3%.

The Fillmore / Dulles 138kV is overloaded to 123.9% of its 228 MVA emergency rating for the loss of the Del Amo / Grand Blanc Generating Station 138kV L. The project contributes 41.8% DFAX (421.6 MW) to this overload.

The Radfords Run Wind Farm / McCormick 138kV is overloaded to 111.5% of its 274 MVA emergency rating for the loss of the Ishua Switch Tower / Boulder Basin 138kV Line. The project contributes 5.1% DFAX (51.2 MW) to this overload.

The Annandale / Centennial 138kV is overloaded to 107.7% of its 223 MVA emergency rating for the loss of the Idylwood Substation / Gallows Rd 138kV. The project contributes 6.8% DFAX (68.1 MW) to this overload.

For the loss of Richland / Dekalb 500kV, the State Street / Robison Park 230kV L loads to 106.6% of its 1,124 MVA rating. Project DFAX: 6.1%.

For the loss of Grand Blanc Generating Sta / Del Amo 138kV, the Idylwood / Gallow Rd Substation 138kV L loads to 105.9% of its 227 MVA rating. Project DFAX: 15.4%.

The Fountain / Thomaston 138kV is overloaded to 105.6% of its 465 MVA emergency rating for the loss of the Grand Blnc Generating Station / Walney 138kV. The project contributes 37.2% DFAX (375.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bull Run / Fillmore 138kV	Rebuild line with higher-capacity conductor	\$7,568,000	100% (\$7,568,000)
Bull Run / Del Amo 138kV	Rebuild line with higher-capacity conductor	\$2,241,000	100% (\$2,241,000)
Fillmore / S Houston 138kV L	Rebuild line with higher-capacity conductor	\$3,473,000	100% (\$3,473,000)
Del Amo / Grad Blanc Generating Station 138kV line	Rebuild line with higher-capacity conductor	\$2,739,000	100% (\$2,739,000)
Castleton Energy Ctr / Sully Substation 138kV line	Rebuild line with higher-capacity conductor	\$2,065,000	75% (\$1,542,000)
Sully / Walney 138kV line	Rebuild line with higher-capacity conductor	\$2,742,000	75% (\$2,047,000)
Grand Blanc Generating Sta / Walney 138kV	Rebuild line with higher-capacity conductor	\$3,201,000	75% (\$2,390,000)
Dessau / S Houston 138kV line	Rebuild line with higher-capacity conductor	\$3,624,000	100% (\$3,615,000)
Gallows Rd / Dessau 138kV L	Rebuild line with higher-capacity conductor	\$2,105,000	100% (\$2,100,000)
Toa Vaca / Thomaston 138kV L	Rebuild line with higher-capacity conductor	\$1,654,000	62% (\$1,024,000)
Fountain / Pleasant Vly Substation 138kV line	Rebuild line with higher-capacity conductor	\$7,000,000	63% (\$4,423,000)
Cub-Run Dp Substation / Pleasant Vly 138kV	Rebuild line with higher-capacity conductor	\$2,280,000	63% (\$1,440,000)
Dulles / Fillmore 138kV	Rebuild line with higher-capacity conductor	\$14,245,000	100% (\$14,245,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Radfords Run Wind Farm / McCormick 138kV	Rebuild line with higher-capacity conductor	\$6,256,000	77% (\$4,793,000)
Annandale Substation / Centennial 138kV	Reconductor line	\$491,000	40% (\$195,000)
State St / Robison Park 230kV Line	Reconductor line	\$3,287,000	9% (\$286,000)
Idylwood / Gallow Rd Substation 138kV L	Reconductor line	\$306,000	87% (\$267,000)
Fountain / Thomaston 138kV line	Reconductor line	\$251,000	63% (\$158,000)
	Total allocated to project		\$54,546,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.